

Problem R28: Viscous Flow in Combined Pipe and Hull System

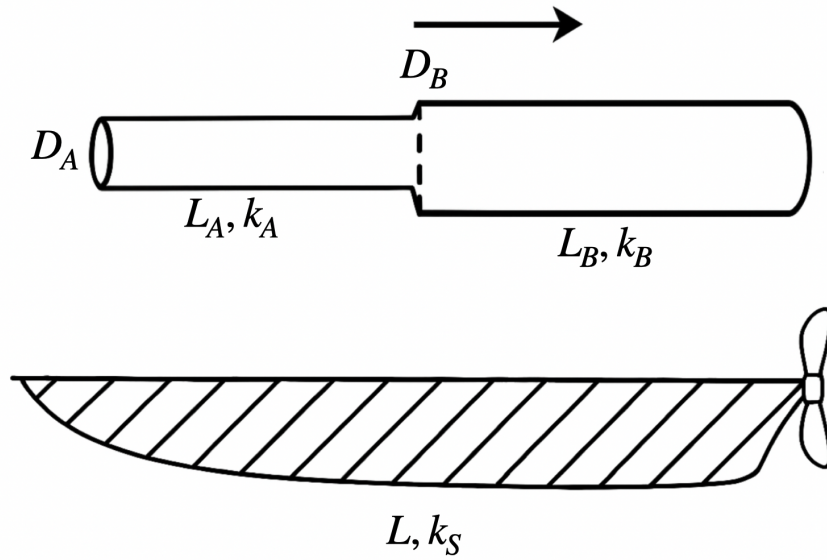


Figure 1: Sketch of the research vessel's hull and propulsion system. The vessel has a hybrid propulsion system with a conventional propeller and a water discharge system through two serial pipe sections.

Scenario

A research vessel uses a hybrid propulsion system composed of:

- a conventional propeller system, and
- a water discharge system that ejects seawater through two **serial** pipe sections (see sketch).

The system is designed such that all pipe flow is **tripped to turbulent regime at the inlet**. The outer hull flow is also assumed **fully turbulent**, avoiding laminar-transition complexity.

Assume seawater at density $\rho = 1025 \text{ kg/m}^3$ and kinematic viscosity $\nu = 1.1 \times 10^{-6} \text{ m}^2/\text{s}$.

Pipe Sections:

- Section A: length $L_A = 30$ m, diameter $D_A = 0.3$ m, roughness $k_A = 0.3$ mm
- Section B: length $L_B = 20$ m, diameter $D_B = 0.2$ m, roughness $k_B = 1.2$ mm

Total volumetric flow rate: $Q = 0.08 \text{ m}^3/\text{s}$

Hull Geometry:

- Length $L = 85$ m
- Draft $T = 6$ m
- Roughness $k_s = 4$ mm

Assume the **wetted area** is approximately $A = 2 \cdot L \cdot T$ (i.e. both vertical sides only).

Ship velocity: $V = 9 \text{ m/s}$

Propeller efficiency: $\eta_S = 0.78$

Solutions

a) Mean velocity:

$$u_A = \frac{Q}{A_A} = \frac{0.08}{\pi(0.15)^2} \approx 1.13 \text{ m/s}, \quad u_B = \frac{Q}{A_B} = \frac{0.08}{\pi(0.1)^2} \approx 2.55 \text{ m/s}$$

b) Reynolds number:

$$Re_A = \frac{u_A D_A}{\nu} \approx \frac{1.13 \cdot 0.3}{1.1 \times 10^{-6}} \approx 3.1 \times 10^5 \text{ (turbulent)}, \quad Re_B \approx 4.6 \times 10^5 \text{ (turbulent)}$$

c) Friction factors (fully rough):

$$\lambda_A = \left(\frac{1}{1.74 - 2 \log_{10}(2 \cdot 0.0003/0.3)} \right)^2 \approx 0.025, \quad \lambda_B \approx 0.035$$

$$\Delta p_V = \sum \left(\lambda_i \frac{L_i}{D_i} \frac{\rho}{2} u_i^2 \right) \approx 0.025 \cdot \frac{30}{0.3} \cdot \frac{1025}{2} (1.13)^2 + 0.035 \cdot \frac{20}{0.2} \cdot \frac{1025}{2} (2.55)^2 \approx 34.6 + 296.8 = 331.4 \text{ Pa}$$

d) Hull friction:

$$c_f = \left(1.89 + 1.62 \cdot \log_{10} \left(\frac{85}{0.004} \right) \right)^{-2.5} \approx 0.0034$$

Wetted area: $A = 2LT = 2 \cdot 85 \cdot 6 = 1020 \text{ m}^2$

$$F = c_f \cdot \frac{1}{2} \rho V^2 A \approx 0.0034 \cdot 0.5 \cdot 1025 \cdot 81 \cdot 1020 \approx 143,950 \text{ N}$$

e) Total power:

$$P = \frac{F \cdot V + Q \cdot \Delta p_V}{\eta_S} = \frac{143950 \cdot 9 + 0.08 \cdot 331.4}{0.78} \approx \frac{1.295 \times 10^6 + 26.5}{0.78} \approx 1.66 \times 10^6 \text{ W} = 1660 \text{ kW}$$